

Pattern Recognition and Neural Networks

Assignment 2

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Phase 1: MLP Baseline & Gradient Flow

1.1 Flattened Temporal MLP (Regression)

Delhi AQI was loaded from `delhi_aqi.csv` after dropping the date column. Each sample uses the previous 72 hours of all 8 features, flattened to a 576-dimensional vector, to predict $\text{PM}_{2.5}$ at $t + 24$.

The implemented regressor is a 3-layer MLP,

$$f(x) = W_3 \phi(W_2 \phi(W_1 x + b_1) + b_2) + b_3,$$

with hidden dimensions 256 and ReLU activations.

The MSE on the test set is **1.1314**.

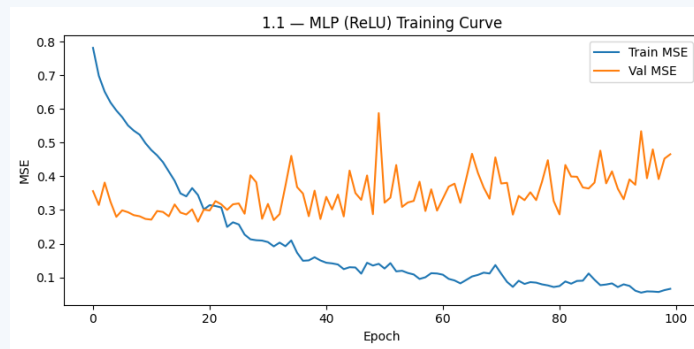


Figure 1: Training and validation MSE for the MLP.

1.2 The Vanishing Gradient Proof

The same architecture was re-initialized with Sigmoid activations and backward hooks were used to record weight-gradient ℓ_2 norms during the first epoch. Over the recorded 205 batches, the mean gradient norms were:

$$\|\nabla W_1\|_2 = 0.262829, \quad \|\nabla W_2\|_2 = 0.886609, \quad \|\nabla W_3\|_2 = 6.380196.$$

The gradient decays in the initial layers. This is mathematically explained by the chain rule of BPTT. For a 3-layer network with Sigmoid activations σ , the gradient of the loss $\mathcal{L}$ with respect to W_1 passes through every activation derivative:

$$\frac{\partial \mathcal{L}}{\partial W_1} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot W_3^\top \sigma'(z_2) \cdot W_2^\top \sigma'(z_1).$$

The Sigmoid derivative satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 0.25$ for all z . Consequently, each additional layer multiplies the gradient by at most $0.25 \times \|W_i\|$. For deep networks this product becomes exponentially small, explaining why $\|\nabla W_1\| \ll \|\nabla W_3\|$.

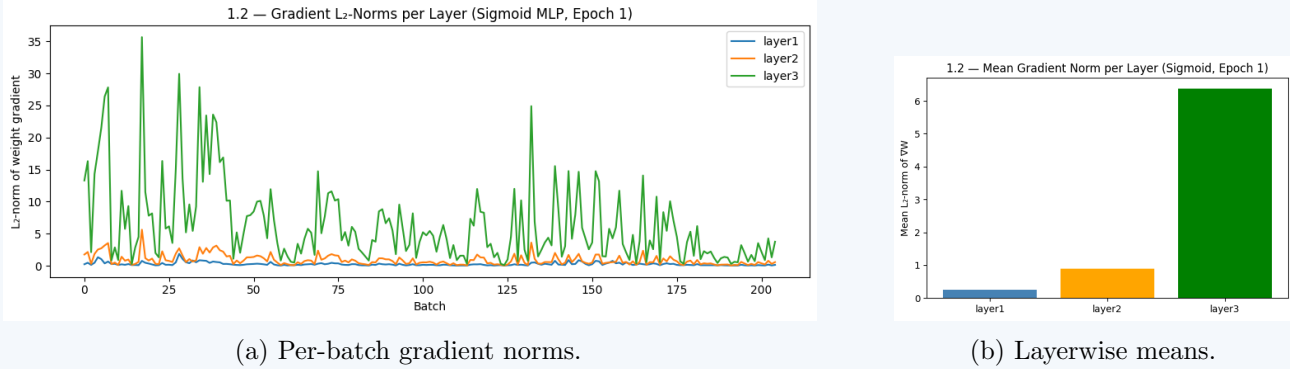


Figure 2: Gradient measurements for the Sigmoid MLP.

1.3 Imbalanced Threshold Classification

The target variable PM2.5 was converted into a binary classification task: 1 if $PM_{2.5} > 200$, else 0. The training split contains 5398 positives out of 13076 samples (41.3%), giving `pos_weight = 1.42`. Two 3-layer classifiers were trained:

- Model A: BCELoss.
- Model B: BCEWithLogitsLoss with `pos_weight`.

The average Precision on the test set was **0.7838** for Model A and **0.7890** for Model B.

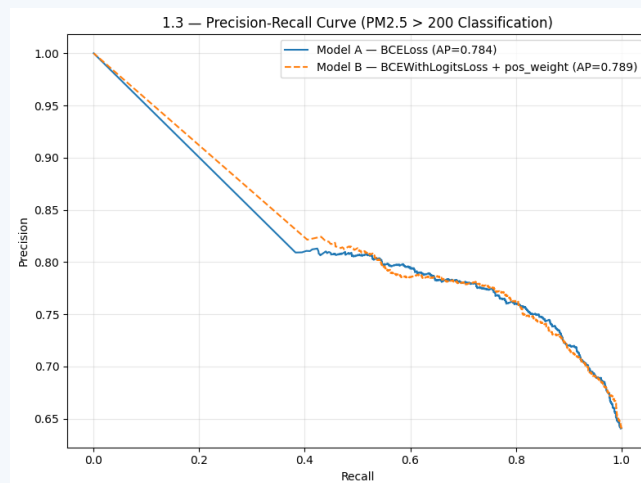


Figure 3: Precision–Recall curves for the two binary classifiers.

Phase 2: CNNs & Spatial Hierarchies

2.4 CNN from Scratch & Receptive Fields

The theoretical receptive field of a single activation in the final feature map is 22×22 **pixels**. The test accuracy: **91.94%**.

Shape of tensor as it passes through each layer:

Layer	Output Shape
Input	[1, 3, 128, 128]
conv1	[1, 8, 128, 128]
bn1	[1, 8, 128, 128]
pool1	[1, 8, 64, 64]
conv2	[1, 16, 64, 64]
bn2	[1, 16, 64, 64]
pool2	[1, 16, 32, 32]
conv3	[1, 32, 32, 32]
bn3	[1, 32, 32, 32]
pool3	[1, 32, 16, 16]
flatten	[1, 8192]
fc1	[1, 256]
fc2	[1, 38]

2.5 CNN Regression Head

Modified the CNN architecture to perform regression:

$$8192 \rightarrow 256 \rightarrow 64 \rightarrow 1,$$

and a final Sigmoid output. The test MAE is **0.1371**.

2.6 Translation Invariance

Accuracy on unshifted test images: **94.00%**.

Accuracy on the same test images after pixel shifting: **90.00%** (a **4.00%** drop).

The reason for this performance drop is because max-pooling provides only local translation invariance. This invariance is limited to shifts smaller than the pooling stride. For a shift of 5 pixels:

- After three max-pool layers each with stride 2, the effective stride is $2^3 = 8$.
- A 5-pixel shift at the input level changes which values land in each pooling window.
- The boundary between pooling windows shifts, so different feature values are selected by the max operation, due to which the activations in the final feature map changes and networks accuracy drops.

Phase 3: RNNs & Sequential Memory

3.7 Vanilla RNN from scratch

Trained the Vanilla RNN model on the 72-hour historical sequences to predict the continuous PM2.5 value.

MSE on the test set: **0.9237**.

3.8 BPTT Decay

Timestamp	Magnitude of the gradient
t = 100	4.71×10^{-1}
t = 50	6.76×10^{-21}
t = 0	0

3.9 Exploding gradients

The largest singular value of the weight matrix is $\sigma_1(W_h) = \mathbf{1.1760} > 1$. This mathematically guarantees exploding gradients as follows. During BPTT, the gradient of the loss with respect to the

hidden state at time step t is

$$\frac{\partial \mathcal{L}}{\partial h_t} = \left(\prod_{k=t}^{T-1} W_h^\top \text{diag}(\tanh'(z_{k+1})) \right) \frac{\partial \mathcal{L}}{\partial h_T}.$$

Since $\text{diag}(\tanh'(z_k)) \preceq I$, the spectral norm of each factor is bounded by $\sigma_1(W_h)$. When $\sigma_1(W_h) > 1$, repeated multiplication grows the product norm like $\sigma_1(W_h)^{T-t}$: for $T - t = 100$ and $\sigma_1 = 1.1760$, this factor reaches $1.1760^{100} \approx 3.05 \times 10^7$, showing that the gradient explodes exponentially in the sequence length.

Phase 4: LSTMs, GRUs & Context

4.10 LSTM Gradient Rescue

Timestamp	Magnitude of the gradient
$t = 100$	7.78×10^{-2}
$t = 50$	3.96×10^{-13}
$t = 0$	0

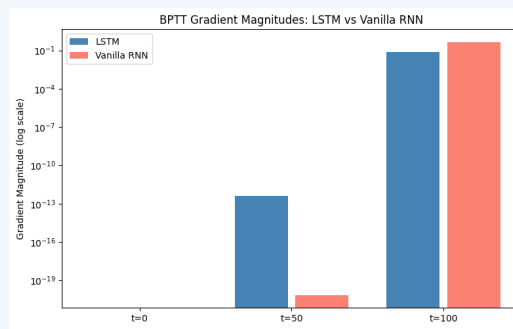


Figure 4: Comparison of the gradient magnitude between RNN and LSTM.

The gradient at $t = 50$ remains much larger in LSTM than in the vanilla RNN.

4.11 GRU vs. LSTM Efficiency

Model	Trainable parameters	Training time per epoch	Test MSE
LSTM	67,201	4.85 s	0.9733
GRU	50,433	8.13 s	0.9702

4.12 Sequence-to-Sequence Forecasting

Modified the LSTM to act as an Encoder-Decoder. The final test MSE is **0.6882**.

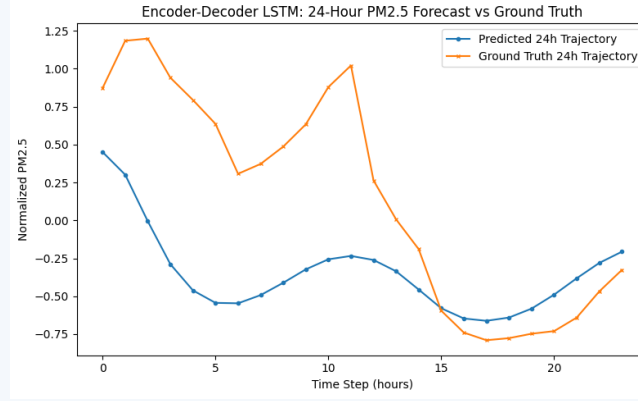


Figure 5: Model's 24-step prediction vs. ground truth.

Phase 5: Transformers & Attention

5.13 Scaled Dot-Product Attention

For tokens $Q, K, V \in \mathbb{R}^{T \times d}$, the scaled-dot product attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

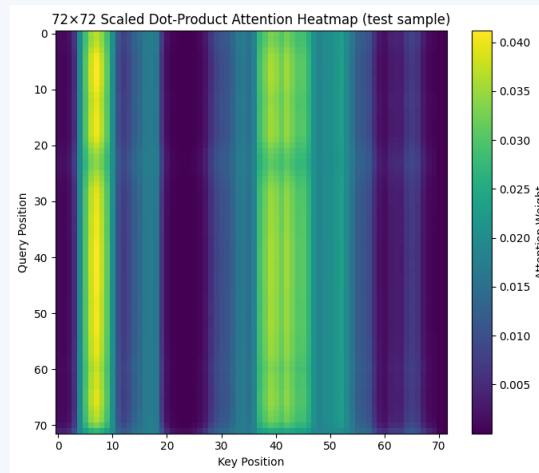


Figure 6: Attention map extracted for one AQI test sample.

5.14 The Necessity of Positional Encoding

Model	Best validation loss	Test MSE
Without positional encoding	0.2517	1.3632
With sinusoidal positional encoding	0.2342	1.0986

The model *without* positional encoding fails to capture temporal trends because self-attention is permutation-equivariant. For a time-series input $X = (x_1, x_2, \dots, x_T)$, any permutation of the tokens produces exactly the same set of output representations, just reordered. Formally, the softmax weights satisfy

$$\alpha_{ij} = \frac{\exp(q_i^\top k_j / \sqrt{d})}{\sum_l \exp(q_i^\top k_l / \sqrt{d})},$$

where q_i, k_j are projections of the token content only — the position of j in the sequence is completely invisible. Hence, the model cannot distinguish whether x_{t+1} comes *after* or *before* x_t .

5.15 Vision Transformer (ViT)

Each 128×128 segmented PlantVillage image is split into 16×16 non-overlapping patches, giving 64 tokens per image, each of dimension $3 \times 16 \times 16 = 768$. The implemented ViT has **64,166** trainable parameters.

Model	Accuracy	Parameters	Parameter Efficiency (Accuracy / parameter)
CNN (Phase 2)	91.94%	2,113,262	4.35×10^{-7}
ViT (Phase 5)	43.29%	64,166	6.75×10^{-6}

Phase 6: Advanced Vision & Imbalance

6.16 Transfer Learning & Freezing

A pretrained ResNet-18 backbone is loaded, the final classification head is modified to support 5 APTOS classes. All pretrained layers are frozen and only the final classification head is trained. Parameter counts for the model:

$$\text{trainable} = 2,565, \quad \text{frozen} = 11,176,512, \quad \text{total} = 11,179,077.$$

On the test split, the model achieves **77.45%** accuracy.

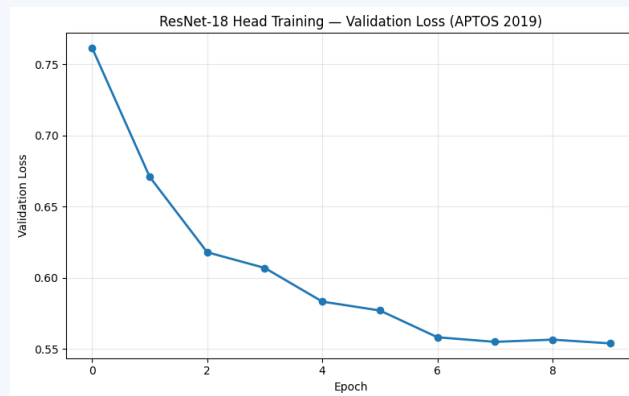


Figure 7: Validation loss curve for the frozen-backbone ResNet-18.

6.17 Ordinal Imbalance Focal Loss

The focal-loss is implemented as

$$\mathcal{L}_{\text{focal}} = -(1 - p_t)^\gamma \log p_t,$$

with $\gamma = 2.0$.

Loss Type	Test loss	Accuracy	Cohen's Quadratic Weighted Kappa
Cross-entropy	0.5753	77.45%	0.7947
Focal loss	0.3159	77.45%	0.8138

The confusion matrices show that the evaluation is strongly class-imbalanced. The focal-loss model improves Cohen's Quadratic Weighted Kappa.

Why Cross-Entropy accuracy is deeply misleading here: The APTOS dataset has severe class imbalance; for example, class 0 constitutes the majority of samples. Mathematically, accuracy is

$$\text{Acc} = \frac{\sum_{c=0}^4 n_{cc}}{N},$$

where n_{cc} is the number of correct predictions for class c and N is the total. Because the denominator N is dominated by the majority class count, improvements on rare classes contribute negligibly to the metric.